



UNIVERSITÀ
DI TRENTO

Frameworks

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Application frameworks

- An application framework is:
a set of tools and functionalities on top of which you can build applications. Thus, it is easier writing an application: the effort of writing all the program code from scratch is taken out
 - A set of tools and functionalities...
- you can choose what pieces of software you need
according the application specifications
- You do not need to use all the features the frameworks offers, you will choose the right parts to use depending on the requirements of your web app

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Application frameworks



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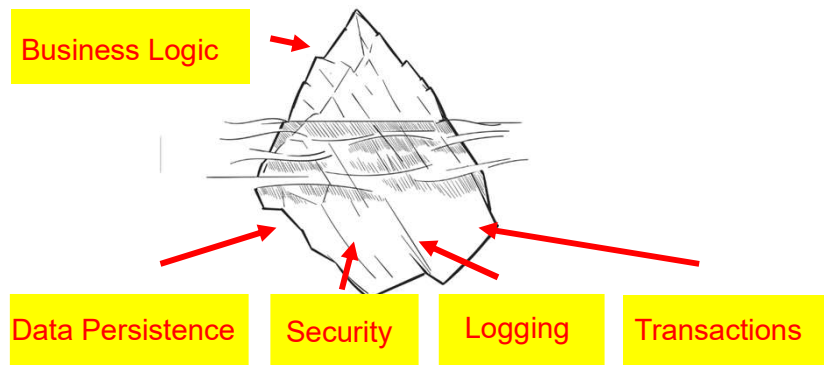
It is an old story

- The idea of framework is not new!
- Everything started by the observation that many apps had similar requirements:
 - Logging error, warning, information
 - Protection mechanisms against common vulnerabilities
 - Communication mechanisms among apps
 - ...
- What is left?

Business logic code implementing the user's expectations.
It is what makes an app different from another one from the functionality point of view

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Functionalities



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Reusing functionalities

- Is efficient to rewrite the same functionalities every time?

NO

- Advantages of the reuse:
 - You spare time and money by reusing code
 - Fewer changes to introduce bugs
 - Support of a community of developers sharing the same code

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But...Pay attention!

- Like everything else in software development, you shouldn't apply a framework in all cases!
- Some examples:
 - You need to have a small footprint (size)
 - Security needs dictate custom code
 - Too much customization makes a framework impractical
 - No benefit switching to a framework

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The Spring ecosystem

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Spring: general information

- Spring it is the most used Java framework today:
 - Open source from 2003
 - It manages infrastructure so that you can focus on application
 - A flexible framework to develops many types of applications
 - Can be used with Java or other JVM languages as Kotlin
 - Current version 7.0.3 (January 2026)
- Spring is modular, you do not need to add the whole Spring to your web app



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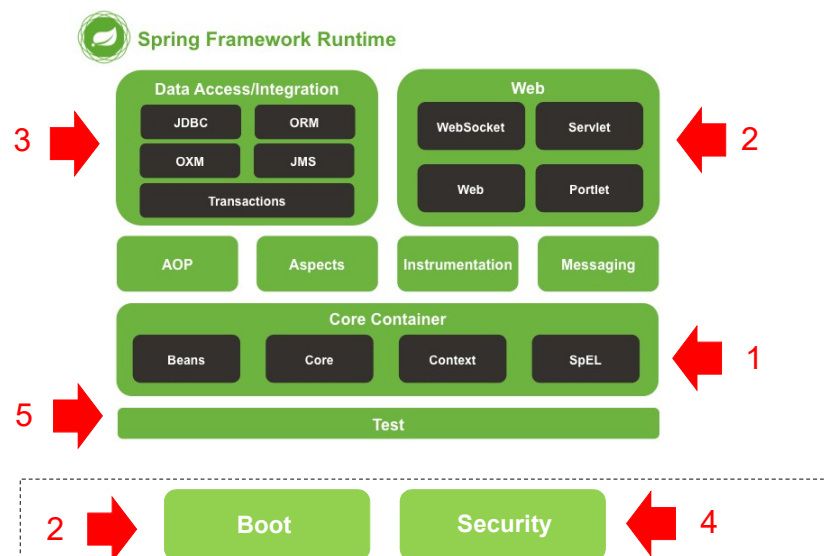
Spring: what you can do

- Common cases in which Spring is a good fit:
 - Development of backend apps
 - Development of desktop apps
 - Development of mobile apps



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Spring (large) architecture



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Companies using the Spring framework



source: <https://www.simplilearn.com/tutorials/java-tutorial/java-frameworks>

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The Spring foundations

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Without IoC

```
double somma (int n, char op){
    double s = 0.0;

    for (int k = 1; k <= n; k++){
        switch (op){
            case 'i':
                s+=inverso(k);
                break;

            case 'q':
                s+=quadrato(k);
                break;

            default:
                break;
        }
    }
    return s;
}
```

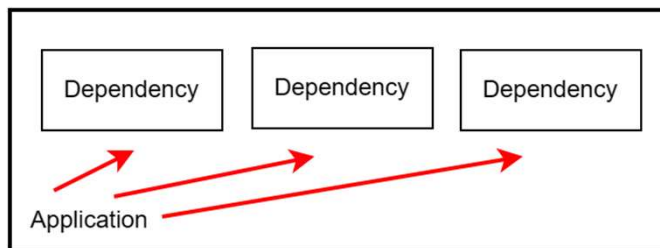
```
double inverso (int a) {
    return 1.0 / a;
}

double quadrato (int a) {
    return double(a * a);
}
```

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Without IoC

WITHOUT IoC



The application executes and controls the dependencies it needs

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Controls and dependencies

```

double somma (int n, char op){
    double s = 0.0;

    for (int k = 1; k <= n; k++){
        switch (op){
            case 'i':
                s+=inverso(k);
                break;

            case 'q':
                s+=quadrato(k);
                break;

            default:
                break;
        }
    }
    return s;
}
  
```

Dependencies

Controls

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Inversion of Control (IoC)

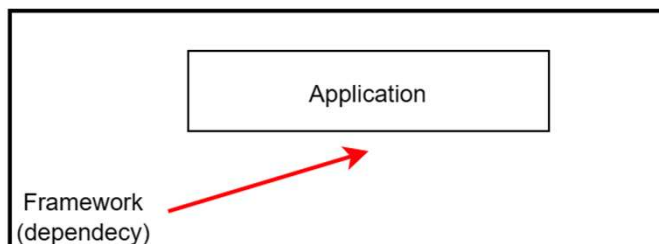
```
double somma(int n, double (*f)(int)) {
    double s = 0.0;
    for (int i = 1; i <= n; i++)
        s += (*f)(i);
    return s;
}
```

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Inversion of Control (IoC)

- In software engineering, inversion of control (IoC) describes a design in which custom-written portions of a computer program receive the flow of control from a generic, reusable external framework

WITH IoC



The application execution is controlled by some other software

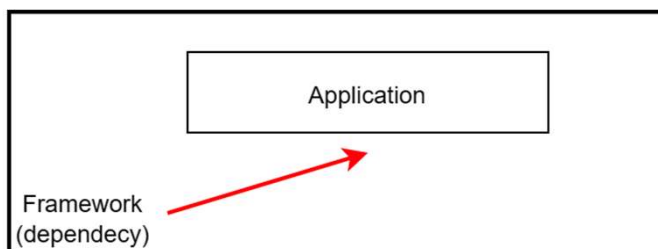
- The key idea behind IoC is that objects do not control their own lifecycle and dependencies, a framework do it

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Inversion of Control (IoC) in Spring

- We give the control to the Spring framework. Through configurations, we instruct Spring on how to manage our code that defines the logic of the application

WITH IoC



The application executes being controlled by some other software

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Advantages of IoC

- Loose Coupling: Components are less dependent on each other, making the system more modular and easier to test.
- Improved Maintainability: Changes to one part of the application (e.g., replacing a dependency) can be done without affecting other parts of the system.
- Flexibility and Scalability: New components can be added or existing ones replaced without changing the underlying structure of the application.
- Centralized Configuration: Configuration of components is handled externally, making it easier to change and manage dependencies.

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A concrete way: Dependency Injection (DI)

- Dependency Injection is a way to make concrete IoC
- Spring injects dependencies at runtime
- How?
 - Constructor Injection: Dependencies are provided to the class via the constructor (preferred method)
 - Setter Injection: Dependencies are set via setter methods after the object is constructed
 - Field Injection: Dependencies are injected directly into fields using annotations

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Aspect Oriented Programming (AOP)

- Object Oriented Programming (OOP) makes it possible writing modular and highly re-usable software using:
 - Classes and interfaces
 - Inheritance, polymorphism, encapsulation
- BUT...there are some horizontal activities called cross-cutting concerns (e.g. authentication, logging, security checks and so on), that cannot be easily isolated but affect multiple components.
- The management of such activities should not be delegated and repeated several times to/by the business logic



Aspect Oriented Programming (AOP) is a complementary paradigm to OOP able to describe and manage such aspects without mixing them with the business logic

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AOP key terminology

- Aspect: a concern to be modularized and applied to various parts of a program. It defines where and when certain behaviors should happen, but it doesn't specify how they should happen.
- Join point: a specific point in the execution of a program where an aspect can be applied
- Advice: the code that is executed when a certain join point is reached. It is what the aspect does at that point

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Advantages of AOP

- Separation of Concerns: Cross-cutting concerns are isolated in separate aspects rather than being scattered throughout the business logic
- Reusability: Since aspects are modular, they can be reused across multiple parts of the application. Thus, code duplication is reduced
- Cleaner Code: By moving cross-cutting concerns out of the core business logic, the code becomes cleaner and more focused on its primary responsibilities
- Improved Maintainability: Changes to cross-cutting concerns can be made just in one place

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Spring Container and Beans

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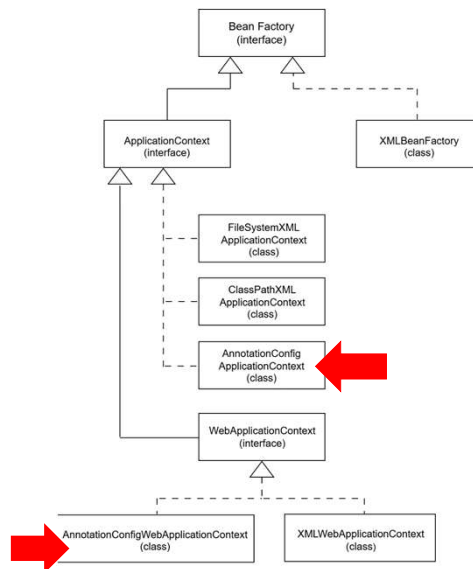
The Spring Container

- The Spring Container is a broad concept of something that:
 - is responsible for managing the objects in an application;
 - handles their lifecycle;
 - is built around the principle of Inversion of Control (IoC), meaning that it controls the flow of the application, not the developer
- A type of Spring Container is the interface `ApplicationContext`. Henceforth, we refer to the Container as `ApplicationContext`

You can think about the `ApplicationContext` as an initially empty box, in which you must put the objects of your web app so that they can work

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The Spring Container



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Spring Beans

- A Spring Bean is a Java object (POJO) whose lifecycle is managed by the ApplicationContext
- Beans can have different roles and scopes
- Among the roles played by Beans we have:
 - Controller
 - Service
 - Repository

A Spring Bean cannot survive outside the Spring framework

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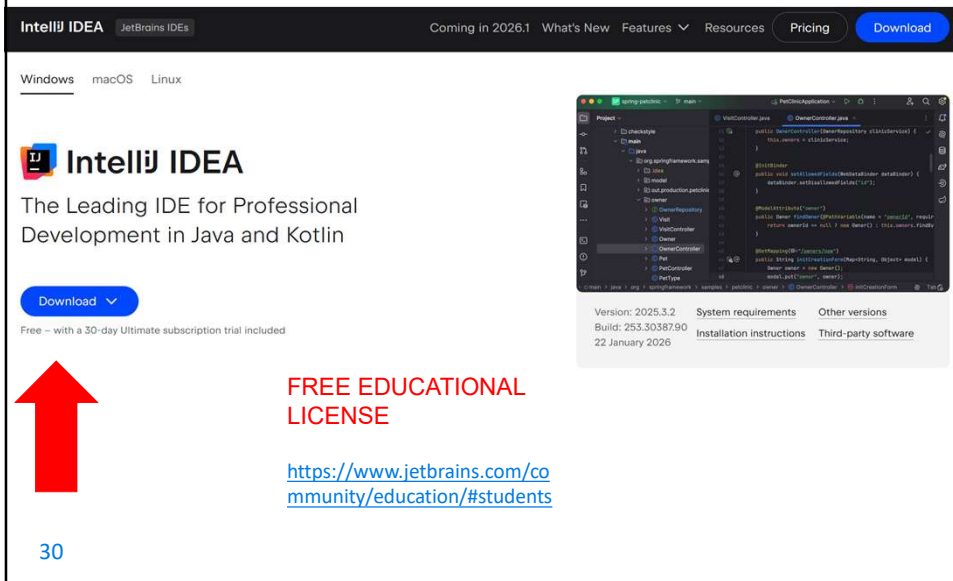
Java Annotations

- Annotations are a powerful mechanism introduced by Java 5 to add metadata to the code
- They provide metadata on a specific code element (e.g. class, method, and so on) without changing its behavior
- Their main purpose is to allow other tools, such as compilers or frameworks, to treat the code in a special way, without the programmer having to write explicit code to achieve that behavior
- They are declared using the symbol `@` followed by the annotation name and possible parameters

Examples: `@Override`, `@Beans`, `@Component`, `@Controller`

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What we need: IntelliJ Ultimate



The screenshot shows the IntelliJ IDEA website. At the top, there's a navigation bar with links like "Coming in 2026.1", "What's New", "Features", "Resources", "Pricing", and a "Download" button. Below the navigation bar, there's a section for "IntelliJ IDEA" with the tagline "The Leading IDE for Professional Development in Java and Kotlin". A "Download" button is prominently displayed. A large red arrow points to this button. Below the button, it says "Free - with a 30-day Ultimate subscription trial included". To the right of the main content, there's a preview of the IntelliJ IDEA interface showing a code editor with Java code. Below the preview, there's a table with links for "System requirements", "Installation instructions", "Other versions", and "Third-party software".

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Java

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JDK 25 JDK 21

Java SE Development Kit 25.0.2 downloads

JDK 25 binaries are free to use in production and free to redistribute, at no cost, under the [Oracle No-Fee Terms and Conditions \(NFTC\)](#).
JDK 25 will receive updates under the NFTC, until September 2028, a year after the release of the next LTS. Subsequent JDK 25 updates will be licensed under the [Java SE OTN License \(OTN\)](#) and production use beyond the [limited free grants](#) of the OTN license will [require a fee](#).
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We need to use JDK25